Semester: 5th
Subject Name:- HPC & Code:- CS 30005
Branch (s): - CSE

AUTUMN **MID SEMESTER EXAMINATION-2025**

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
High Performance Computing
[**CS30005**]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give theiranswers in their own words as far as practicable and all parts of a question should beanswered at one place only.

1. Answer all the questions. [1 Mark X 5]

a) A 4 stage pipeline has stage delays as 100, 80, 150,120 ns respectively. There is a latch delay of 5ns each. Calculate the pipeline cycle time and non-pipeline cycle time?
Ans: Pipeline cycle time=155ns, Non-pipeline cycle time=450ns

b) Explain how the data forwarding techniques is useful in reducing the data hazard.
Ans: The output of ALU is feed forward to input of ALU. Also the output of MEM stage is feed forward to the input of ALU.

c) In an ideal pipeline with 'k' number stages and 'n' number of instruction, the maximum speedup can be achieved upto _____ under _____ condition?
Ans: (i) k (ii) n is very large or n tends to infinity

d) Assume that 20 % of your program is floating point operations and rest of the program is integer operations. A design team D1 proposes a speed up for the floating point operations by five times using a dedicated floating point hardware circuit. Another design team D2 proposes the speed up for integer operations by two times. Which design has the highest performance?

Ans: D2 has the highest performance.

D1:	D2:
Speedup=1/(0.8+(0.2/5))=1.19	Speedup=1/(0.2+(0.8/2))=1.66

e) Write the name of the different solutions that are available for control hazard.

Ans: 4 solutions available for control hazard.

(i) flush pipeline, (ii) predict branch not taken, (iii) predict branch taken, (iv) delayed slot

2. (a) Write the machine code in hexadecimal format for the following assembly language MIPS instructions. Assume the function code for the DADD is 20 and opcode for the DSUB is 30. [2 Marks]

DADD R10, R12, R15

DSUB R5, R7, #10

Ans:

DADD R10, R12, R15

Machine code format: 000000 01100 01111 01010 00000 010100

Hexadecimal format: 018F5014

DSUB R5, R7, #10

Machine code format: 011110 00111 00101 0000 0000 0000 1010

Hexadecimal format: 78E5000A

(b) Discuss Instruction Set Architecture (ISA) and mention all seven dimensions to be considered while designing an ISA. [3 Marks]

Ans: The Instruction Set Architecture (ISA) is the part of the processor that is visible to the programmer or compiler writer. The ISA serves as the boundary between software and hardware.

7 dimesion of ISA:

(i) Class of ISA, (ii) Memory Addressing, (iii) Addressing modes, (iv) Types and size of operands, (v) Operations, (vi) Control flow instructions (vii) Encoding an ISA

3. (a) Derive the expression for speedup of an ideal pipeline with respect to 'k': no. of stages, 'n': no. of instructions, 't': stage delay and there is no latch delay. [2 Marks]

Ans: *k-stage* pipeline processes *n* tasks in k + (n-1) clock cycles:

Total time to process *n* tasks in pipeline

$$T_k = [k + (n-1)]\ t$$

For the non-pipelined processor

$$T_1 = n\ k\ t$$

Speedup= $T_1 / T_k = n\ k / [k + (n-1)]$

(b) Consider a nonpipelined processor in which ALU instruction, branch instruction and memory operations take 5nsec, 7nsec and 9nsec respectively. Assume the relative frequencies of these operations are 50%, 10% and 40% respectively. Assume the cycle time is 1nsec. Compute the speedup due to pipelining (assume the pipeline overhead is 0.2nsec and ignore the effects of branching) [3 Marks]

Ans:CPI for nonpipelined=0.5*5+0.1*7+0.4*9=6.8nsec

CPI for pipelined=1+0.2=1.2nsec

Speedup=6.8/1.2=5.6

4. (a) Explain the different solutions that are available to the data hazard. [2 Marks]

Ans: Solutions that are available to the data hazard

(i)Stalling, (ii) Forwarding, (iii) Instruction Reordering

(b) Find out the total no of clock cycles required to execute the following instructions without and with operand forwarding? [3 Marks]

I1: LD R_1, 100(R_2)

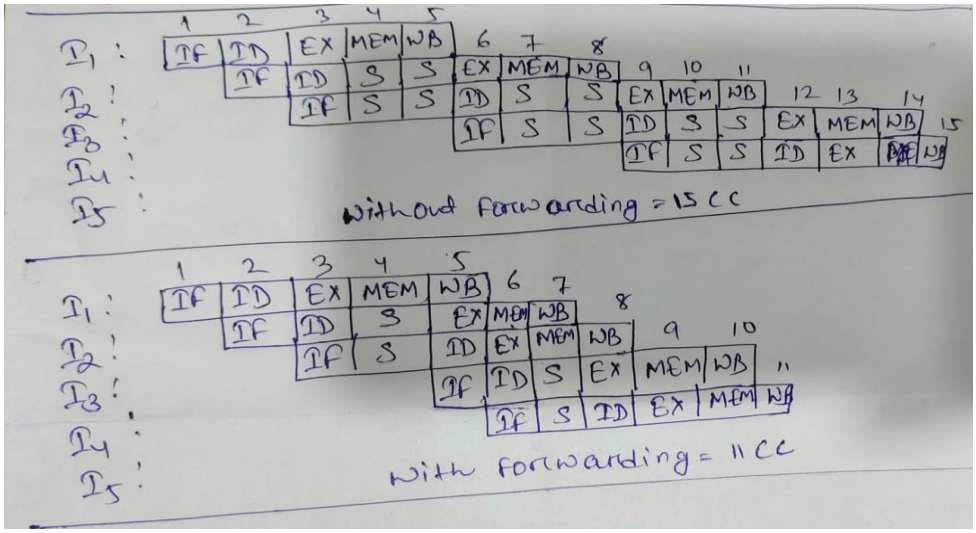
I2: DSUB R_3,R_1,#20

I3: LD R_4,100(R_3)

I4: SD R_5, 10(R_4)

I5: DADDIU R_2,R_3,#4

Ans:



5. (a) Find out dependencies (RAW, WAR, and WAW) for the following MIPS instructions .

I1: DADD R_1,R_2,R_3

I2: LD R_4,0(R_1)

I3: DADDUI R_5, R_4,#10

I4: SD R_4,0(R_5) [2 Marks]

Ans:

RAW: (I1, I2), (I2, I3), (I3, I4), (I1, I3), (I2, I4), (I1, I4),

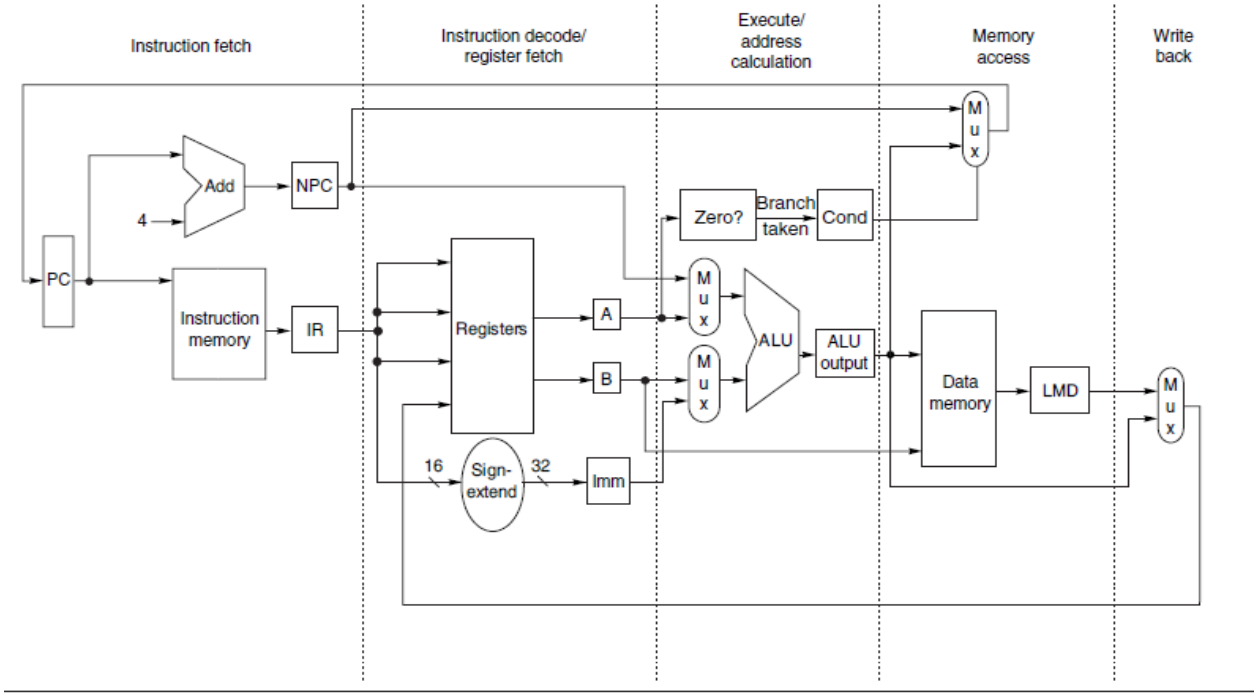
WAR: NO

WAW: NO

(b) Draw a neat diagram for MIPS Datapath and specify the different registers in each segment of MIPS Pipeline. [3 Marks]

Ans:

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Register names in each stages:

IF Stage:	ID Stage:	EX Stage:	MEM Stage:	WB Stage:
PC, NPC, IR	A, B, Imm	Cond, ALUouput	LMD	Nil

*** Best of Luck ***